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Section	**AD1**
Subeject	**AI Lab**
Lab work	**week 2**

Task 1:

Code :

```
num = int(input("Enter a number: "))
is_prime = True

if num <= 1:
    is_prime = False
else:
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

if is_prime:
    print("Number is Prime")
else:
    print("Number is Not Prime")
```

output:

[2]
5s

```python
num = int(input("Enter a number: "))
is_prime = True

if num <= 1:
    is_prime = False
else:
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

if is_prime:
    print("Number is Prime")
else:
    print("Number is Not Prime")
```

```
Enter a number: 4
Number is Not Prime
```

```
num = int(input("Enter a number: "))
is_prime = True

if num <= 1:
    is_prime = False
else:
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

if is_prime:
    print("Number is Prime")
else:
    print("Number is Not Prime")
```

```
Enter a number: 7
Number is Prime
```

Task 2:

Code:

```
def factorial(n):
    result = 1
    for i in range(1, n+1):
        result = result * i
    return result


num = int(input("Enter a number: "))
print("Factorial is:", factorial(num))
```

output:

```
16]
18s
def factorial(n):
    result = 1
    for i in range(1, n+1):
        result = result * i
    return result

num = int(input("Enter a number: "))
print("Factorial is:", factorial(num))

Enter a number: 7
Factorial is: 5040
```

Task 3:

Code:

```
text = input("enter voweles in string ")

vowels="aeiouAEIOU"

count= 0

for ch in text:

  if ch in vowels:

    count +=1

print("number of vowels ",count)
```

output:

```
[21]
19s
text = input("enter voweles in string ")
vowels="aeiouAEIOU"
count= 0
for ch in text:
  if ch in vowels:
    count +=1
print("number of vowels ",count)

enter voweles in string khuram sofii ilyas
number of vowels  7
```

Task 4:

Code :

```
numbers=[1,2,3,3,4,7,5,8,9,0,2,1,12,34,56]

print("the even numbers are ")

for num in numbers:

  if num % 2==0:

    print(num)
```

output:

```
[24]
✓ 0s
numbers=[1,2,3,3,4,7,5,8,9,0,2,1,12,34,56]
print("the even numbers are ")
for num in numbers:
  if num % 2==0:
    print(num)

the even numbers are
2
4
8
0
2
12
34
56
```

Task 5:

Code :

```
student = {
   "Name": "Ali",
   "Roll No": 101,
   "Department": "Computer Science",
   "Marks": 85
}
print("Student Information")
print("Name:", student["Name"])
print("Roll No:", student["Roll No"])
print("Department:", student["Department"])
print("Marks:", student["Marks"])
```

output:

```
[33]
0s
student = {"Name": "khuram","Roll No": 38,"Department": "Artifical intaligance","Marks": 77}

print("Student Information")
print("Name:", student["Name"])
print("Roll No:", student["Roll No"])
print("Department:", student["Department"])
print("Marks:", student["Marks"])

Student Information
Name: khuram
Roll No: 38
Department: Artifical intaligance
Marks: 77
```

Task 6:

Code:

numbers = [1,2,8,3,2,4,1,5,3,6,2,3,7,9,5]

unique_numbers = list(set(numbers))

print("Original List:", numbers)

print("after removing duplicates:", unique_numbers)

output:

```
[36]
0s
numbers = [1,2,8,3,2,4,1,5,3,6,2,3,7,9,5]

unique_numbers = list(set(numbers))

print("Original List:", numbers)
print("after removing duplicates:", unique_numbers)

Original List: [1, 2, 8, 3, 2, 4, 1, 5, 3, 6, 2, 3, 7, 9, 5]
after removing duplicates: [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Task 7:

Code :

```
students = (
    (1, "khuram", (78, 85, 83)),
    (2, "miksi", (79, 86, 90)),
    (3, "soffi", (71, 84, 77)),
    (4, "ali", (60, 69, 76)),
    (5, "zaki", (58, 65, 81))
)
highest_avg = 0
top_students = []
count_above_75 = 0
for student in students:
    roll, name, marks = student
    avg = sum(marks) / 3
    print(name, "average is :", avg)
    if avg > highest_avg:
        highest_avg = avg
        top_students = [name]
    elif avg == highest_avg:
        top_students.append(name)
    if avg > 75:
        count_above_75 += 1
print("highest average students is :", top_students)
print("students above 75 average is :", count_above_75)
```

```
updated_students = (
  (1, "khuram", (78, 85, 83)),
   (2, "miksi", (79, 86, 90)),
   (3, "soffi", (71, 84, 77)),
   (4, "ali", (60, 69, 76)),
   (5, "zaki", (58, 65, 81))
)
print("the updated records is :", updated_students)
```

output:

```
            top_students = [name]
        elif avg == highest_avg:
            top_students.append(name)
        if avg > 75:
            count_above_75 += 1
    print("highest average students is :", top_students)
    print("students above 75 average is :", count_above_75)
    updated_students = (
       (1, "khuram", (78, 85, 83)),
        (2, "miksi", (79, 86, 90)),
        (3, "soffi", (71, 84, 77)),
        (4, "ali", (60, 69, 76)),
        (5, "zaki", (58, 65, 81))
    )
    print("the updated records is :", updated_students)

khuram average is : 82.0
miksi average is : 85.0
soffi average is : 77.33333333333333
ali average is : 68.33333333333333
zaki average is : 68.0
highest average students is : ['miksi']
students above 75 average is : 3
the updated records is : ((1, 'khuram', (78, 85, 83)), (2, 'miksi', (79, 86, 90)), (3, 'soffi', (71, 84, 77)), (4, 'ali', (60, 69,
```

Task 8:

1. Difference between while and for loop

For loop: Used when we know how many times the loop will run.

While loop: Used when the loop runs until a condition becomes false.

Example:
for loop runs a fixed number of times, but while loop depends on a condition.

2. What is Function Modularity?

Function modularity means **dividing a big program into small functions**.
Each function does **one specific task**.
This makes the program **easy to understand and manage**.

3. Difference between List and Tuple

List: Can be changed (add, remove, update elements). Uses **[ ]** brackets.

Tuple: Cannot be changed after creation. Uses **( )** brackets.

4. Why are Sets used?

Sets are used to **store unique values only**.
They **automatically remove duplicate values** from data.

5. What is Dictionary Key Restriction?

Dictionary keys must:

Be **unique** (no duplicates).

Be **immutable** like **strings, numbers, or tuples**.

Example:
{"name": "Ali", "age": 20}